



Module 21

Race Preparation & Execution

Race day represents the culmination of all of your client's hard work. As such, there should be no surprises on race day and a strategy should be in place to give your client the best chance of hitting his or her goal. This module discusses the areas and aspects necessary to ensure that your client has a positive experience on race day!

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- General preparations
 - Course knowledge
 - Weather
 - Prepare for the unexpected
- Rules
- Areas of focus the week before a race
- Areas of focus the day of a race
- Race-day schedule and logistics

❖ INTRODUCTION

Your clients might go through a wide range of emotions on race day, including being excited, nervous, or even scared. This is natural, and to some degree, it would be unnatural not to have anxiety the day of an event. It is your responsibility to remind clients that they have trained for this day and they should feel confident in their preparation. Remind them to stick to the game plan and to have fun.

Four things need to be focused on in the weeks leading up to an event:

- Assure clients that they have prepared properly.
- Have clients write down a schedule for race day.
 - Make sure they go over the race-day checklist to ensure they have everything they will need for the event.
- Ensure that they have clearly read and understand the rules, directions, and schedule of events (e.g., race number pickup).
- Have them mentally go over the strategy that you have laid out for them the morning of the event.

❖ GENERAL PREPARATIONS

COURSE KNOWLEDGE

The course has a substantial influence on how difficult a running race is and, more specifically, how a race is trained for. Prior knowledge of a course will tell athletes where challenging parts of the course are, locations of water stops, and where they can possibly conserve energy. As noted in the *Pacing* module, races often provide course and elevation maps. This is very helpful in the design of a training program as well as for pacing and race strategies. Another way to gather information about a race course is to speak to those who have done the race before. Online blogs and message boards are also good sources of event information, assuming that it is not the first year it's being held or that the course hasn't changed. Some of the things to focus on are:

- Notable climbs and descents

- Road conditions
- Final few miles and especially the final mile
- Location of aid stations/bathrooms

Tangent Lines

A tangent line is a straight line that touches a curve at a point. By following “tangent lines,” a runner is able to run a course in the most efficient way possible.

This is done by making the course as “small” and “fast” as possible. This is accomplished by running the shortest distance between point A and point B. This tactic is called *running the tangents*. Course distances are typically measured at the tangents of turns, so by not racing the tangents a runner is effectively racing farther than the course actually is. *Racing the tangents* is the best way to decrease one’s race time without expending extra physical effort.

In figure 21.1, the blue lines represent the road and the red line represents the path of a runner. This illustrates how racing the tangents can substantially “shorten” a course. This is possible only on a race course that is closed to traffic.

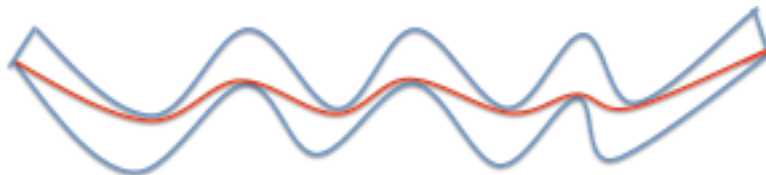


Figure 21.1 Racing the tangents

BE AWARE!

Running the tangents throughout an entire course is difficult, if not impossible, when there are a lot of participants. Therefore in large, crowded races, your client should never cut someone or a group of people off to run a tangent line, as this is dangerous and impolite.

Google® Maps

If a race provides a course map, you can utilize the street view function of Google® Maps to view the course. This is a great tool as you can virtually “drive” the entire course. Some of the aspects of a course that can be ascertained are:

- Hills and descents (steepness)
- Landmarks (ex: one mile to go there is a red barn on the left of the road)
- Turns

Be aware that if you are using the map for landmarks, depending on when the mapping was done, the landmarks might have changed (ex: red barn was torn down).

WEATHER

The weather influences race strategy as well as race preparation. The temperature affects many things, including apparel, hydration strategy, and effort level. While it is important to have trained in the expected weather conditions of an event, your client must be prepared for unforeseen temperature fluctuations.

Rain and inclement weather do not impact just performance, but also the overall safety of your client. Strong winds also affect a client's overall time.

In order to prepare for bad weather, both physically and mentally, one must train in it. It goes without saying that any training done in bad weather must also be done in a safe manner.

For clients who are competing or training in extreme heat, heat acclimatization is necessary. Information on this is provided in the *Illness and Injury* module.

PREPARE FOR THE UNEXPECTED

While there will always be unexpected issues that come up on race day, it is your responsibility to identify and adapt a client to as many potential issues that can occur on race day during training. This serves three primary purposes:

1. Reduce the anxiety of a client by building his or her confidence
2. Safety
3. Increase performance

Potential Issues

- Bad weather: rain, wind, extremely hot/cold temperatures
- New course
- Adjusted start times
- Sickness
- Body pains (e.g., knee, shin, cramps, hamstring)
- Injury
- Food/drink offerings at rest stops are not the same as noted in the race information
- Extreme fatigue
- Tripping/falling
- Blisters/chafing
- Getting lost
- GPS watch malfunction

It is important that you identify potential issues as well as possible solutions – or at least ways to mitigate the severity of them. In addition to the solutions discussed in this module, it is advantageous to speak to other coaches to get additional resources on how to minimize the impact of unexpected issues that can pop up on race day.

Common Issues

Of the aforementioned issues, some of the more common ones that a runner may encounter and experience on race day are listed below. It is imperative that as a coach you explain the issue and how to minimize the impact.

- Weather (e.g., rain, cold, heat)
- Sickness
- Cramps
- Injury

Weather

Your clients should keep an eye on the forecast. Regardless of the forecast, it is advised that clients pack a wide assortment of clothes so they are ready for any weather condition.

Sickness

Clients who are sick should not race, and if they are in the race already, they should not continue.

Exercise-Related Cramps

If a cramp is localized and can be treated via stretching and/or self-massage, then it is fine to continue. However, once cramps become so severe that rest, self-massage, and stretching do not eliminate the cramping, it is advisable to stop. Cramping is typically associated with the following:

- Running at a high-intensity level
- Running for longer than having previously trained for
- Low glycogen/electrolyte stores*
- Dehydration*

* Most current research has debunked the myth that low glycogen and mild to moderate dehydration levels cause muscle cramps (763-765).

Cramping, and more specifically severe cramping, is typically preventable.

Injury

Through proper training, many injuries can be prevented. However, if an injury does occur during an event, clients need to be able to assess accurately if they should or should not continue. Reasons not to continue are:

- Injury will get worse if they continue
- Head or back injury
- Excessive pain
- Suspected ligament or tendon injury
- Potential for infection
- Broken bone

Clients who are unsure if they should continue in an event should seek out medical advice at an aid station, if possible.

RULES

Part of having a successful strategy is having a clear understanding of the rules. It is the responsibility of both you and your client to fully understand and comply with the rules of an event and that of the race's sanctioning body, if applicable. The rules of an event should be posted on the event's website.

WEEK BEFORE THE RACE

Following are areas to focus on the week before a race (most applicable to half and full marathons and ultra-marathons):

1. Rest and Recovery
2. Nutrition and Hydration
3. Supplies
4. Confirm Travel Plans (if any)
5. Athlete Tracking
6. Mental Preparation

Rest and Recovery

Conserving energy must be a primary focus before a race. This is an all-encompassing focus. Aside from any workout sessions, excess energy expenditure should be avoided or kept to a minimum. The longer the race, the more this applies.

Nutrition and Hydration

Eating healthy, familiar foods is important at all times during the training process but especially the week prior to an event. A client must place a large focus on staying well hydrated. Refer to the *Sports Nutrition* module for more information on nutrition and hydration.

Supplies

Anything and everything needed for an event must be sourced and organized prior to or during this week. Following is a list of items/supplies that clients should reference to make sure they are prepared for their event:

- Nipple covers for men, if chafing is a concern
- Running shoes
- Socks
- Anti-chafing balm such as *Body Glide*®
- Running hat/visor
- Running clothing (i.e., shorts, shirt, tights, sports bra, etc.)
- Sunglasses
- Compression socks, if used
- Throwaway clothes if it's a cold-weather event

General

- Pre-/post-race food
- On-the-run food/drinks
- Dry clothing and comfortable shoes for after the race
- Waterproof sunscreen
- Photo ID/passport
- Towels
- Toilet paper
- Medical needs (e.g., insulin, inhaler, etc.)
- Heart rate/GPS watch
- Watch/alarm clock
- Directions to race
- Camera
- Wallet/money
- Phone
- Reservation information (if any) for lodging, restaurants, transportation
- Race information, race number/timing chip/confirmation of race entry
- Rain gear
- Extra running clothing
- Hydration sources
- Emergency contact information

If your client is flying to an event, running shoes *must* be in his or her carry-on.

Running shoes are very specific to an individual and can spell disaster if they are lost in transit. This is not to say other equipment/apparel is not important. However, running shoes are critical.

Confirm Travel Plans

If traveling to an event, it is suggested that your client get confirmation of any and all travel arrangements such as plane, hotel, restaurant, and car rental reservations.

Athlete Tracking

Many large half and full marathons have athlete tracking services that follow the midrace progress and finish time of runners. This is facilitated by participants running over timing strips that capture their time. If family and friends enter a participant's race number in the athlete tracker (typically a mobile phone app), they will get updates sent directly to their phone.

Note: As glitches with tracking apps and timing strips can occur, runners should inform their friends and family not to worry if they don't get midrace splits.

❖ DAY BEFORE THE RACE

There should be three areas of focus the day before a race.

1. Take care of all event preparations

This includes all operational items such as race number pickup and pickup any last-minute items needed for race day such as food, hydration, and throwaway clothes.

2. Proper fueling strategy

This should have been practiced during training. Aside from meals, it is a good idea for your client to bring along snacks that are familiar and healthy. Depending on the size of a race, the host town or city may be overrun with hungry runners looking for food. This means hundreds – if not thousands – of runners and their families/friends all vying for the same places to dine. Therefore, advanced planning such as reservations (months ahead) is advisable. Also, hydrate, hydrate, hydrate!

3. Rest

It's tempting to walk around the expo area or check out the sights, especially if a race is in a fun area and your client is with friends and family. However, you must remind clients that they have trained very hard for their race and that they must conserve their energy for the next day. More specifically, your clients should keep walking around to a minimum and get enough sleep.

If clients want to go sightseeing and do tourist things, they are best done after the event is over and ideally one or two days after the event. The longer the event, the more recovery time is needed.

AT EVENT: DAY BEFORE EVENT

The race schedule will inform clients where they need to be and at what time. Below are some things to consider:

Athlete Check-In

This is where all registered athletes go to confirm participation in the event. At this point, athletes will usually get a race packet with all the information pertaining to the event. For larger events, this is often at the same location as the race expo. It is a good idea for your client to have the following items on hand:

- Photo ID
- Confirmation of entry (i.e., printed copy of race entry confirmation)

Expo

Expositions are commonplace at races, big and small. Some are held on-site at the race course while others are “off-site” (often in the host hotel or a convention hall). Expos are great for checking out new gear and present a good opportunity to pick up fueling samples to use at a later time.

Remember: *Nothing new on race day!* Race day is when your client uses what is tried and tested during training so that everything is familiar and safe.

Pasta Dinner

Many events have a pasta dinner the evening before the race for participants and family members. These are great to attend because they are often cheaper than dining at restaurants. They are typically held at the host hotel or somewhere nearby. It's also a great time for your client to meet fellow competitors. The one potential downside is that your client typically has no control over what food is available, how it is prepared, or the exact content. For example, if your client has special dietary needs, this might not be the best idea. In addition, the times when the pasta dinner is offered might not fit in with your client's fueling schedule.

❖ RACE DAY

Race day, and the days leading up to a race, can be a very intense, nerve-racking period. This is typically magnified for new runners. Through stomach butterflies and nervous energy, it is easy for your client to get thrown off track and lose focus of what needs to be done prior to the race start. For this reason, it is important for them to write down a race-day timeline. Once clients get the race schedule that notes where and when things need to be done (e.g., race start time, shuttle bus, deadline to be in start corrals), they should create a race-day timeline. This will give clients confidence that they are not forgetting anything and they are where they need to be at particular times. A sample race-day event timeline might look like this:

- 4:45 a.m.:** Wake up
- 5:15 a.m.:** Breakfast
- 6:15 a.m.:** Get changed into running clothing and throwaway clothing
- 7:30 a.m.:** Check out of hotel
- 8:00 a.m.:** Eat prerace food

8:15 a.m.: Bathroom break
8:30 a.m.: Warm up
9:00 a.m.: RACE START

Wake Up!

The months of training and sacrifices all culminate on race day. So what is the most important thing to remember? The alarm clock!

Whether it be oversleeping an alarm or forgetting to set one, the end result is the same: missing the event. Therefore, it is in your client's best interest to set multiple alarms and, if available, arrange for a wake-up call. There are many reasons for a DNF. However, not making the start line should not be one of them.

Eat

Regardless of the length of an event, your client needs to be fueled appropriately. The timeline of when to eat and what to eat should be worked through in training to ensure ample time to eat and digest food without negative consequences. The longer the event, the more critical prerace fueling is.

Warm Up

For shorter-distance events, a proper warm-up is an important part of race day. Physiologically speaking, a warm-up has many positive effects on the body (261, 262).

- Warm blood delivers more oxygen to the muscles
- Dilates blood vessels, which are constricted during rest
- Muscles contract faster and with greater force when warm versus "cold"

For short-distance events, the minimum amount of time spent warming up should be five minutes and the maximum 30 minutes.

Like all other aspects of racing, a warm-up should be practiced in training to determine the optimal duration and intensity. A warm-up is meant to enhance performance and decrease the chance of injury, not to prefatigue an athlete.

RACE-DAY LOGISTICS

Timing Chip

This is typically included in the registration packet. The chip is typically attached to the race number or is embedded on a strap that gets fastened around shoelaces. As this chip records an individual's time, a client must know where it is on race day. No chip = no official race time.

Last Bathroom Stop

Increased anxiety often equates to increased bathroom stops. The port-a-potty area is often very hectic prior to the start of a race. As such, lines can be long. Clients should factor in time to use a port-a-potty prior to the race. Additionally, as there is a better-than-average chance the toilet paper will be gone, your client should have a personal supply!

The Start

Starting areas of large events will likely be divided into corrals and/or waves based on participants' estimated paces – fastest up front, slowest in the back. The corrals are often denoted by the color of the race number bibs and also by the race numbers. The event schedule will identify when and where your client will start.

Prior to the start, have your clients replay the race game plan in their mind – nothing should come as a surprise. Having a game plan will help to reduce prerace anxiety and increase performance.

If a client must be at the start line long before the gun goes off and cold is expected, it is advised to wear warm clothes that can be discarded before the start. These are often termed *throwaway clothes*, and many events donate the garments left behind by runners. Thrift stores are great places to pick up cheap, warm clothes!

During the Race

By the time the start gun goes off, your client should know the pacing and fueling strategy, as well as how to deal with most unexpected things that might come up. Other tactics, such as *running the tangents*, need to be remembered and implemented on race day.

The Finish

If clients will have family/friends there to cheer them on, they should mutually agree upon a location to meet at. Some events do not allow spectators in the finish area, so selecting a predetermined meeting location is important. Large events often have a *family and friends area* near the finish line. Additionally, directly after a race, it is advised that clients walk around to prevent blood pooling in their legs.

Post-Race Fueling

The event is over and it's time to celebrate ... almost. Your client will need to refuel and hydrate at the end of the event, either using the food and drinks made available by the event or brought by the individual's family, friends, or coach.

Completing a running race is a huge accomplishment and deserves celebrating, but not at the expense of one's well-being. It is advisable for your client to wait a few hours after the race and preferably at dinner that evening to have alcohol – and only then in moderation. Alcohol has a much greater effect on an athlete post-race than under normal conditions. The primary post-race drink should be water and/or an electrolyte replacement drink.

Runners sacrifice a lot to train for a race. Some of these things are their free time, as well as their favorite (and maybe not so healthy) food and drinks. Because of these sacrifices, many runners go crazy at dinner after finishing a race to reward themselves for all of their sacrifices. While this is certainly deserved, the body is probably not used to a triple-decker cheeseburger with a plate of gravy fries. Therefore, it is advisable to eat relatively healthy and in moderation.

Post-Race Care

Many events have complimentary post-race massages and stretches available for participants. This is a service that will help in the recovery process. While hopefully your client is not a candidate for the medical unit at an event, it is important for your client to know that there is that level of support, if need be.

TO ICE OR NOT TO ICE?

Depending on when clients are leaving an event, they may or may not have access to a hotel room. If they do have access to a hotel or friend's house, a short (10 minutes or less) ice bath may be helpful. This type of treatment is called *cryotherapy* (cold therapy) and works to reduce inflammation.

There have been multiple studies on this topic in regard to its validity in respect to its effect on recovery and inflammation reduction (766–769). The results are mixed. Following are the main points from the most current research in this area:

1. Cryotherapy reduces inflammation. However, this may or may not correlate with enhancing recovery time. In fact, some research argues that it extends recovery time.
2. Most individuals who undergo cryotherapy feel better and more recovered. Without meaningful physiological evidence to show improved markers for recovery, the feeling of recovery is thought to be largely psychological. The **placebo effect** is discussed in greater detail in the *Mental Training* module.
3. Aside from the potential to delay recovery, there is little to no danger to cryotherapy so long as the water is not too cold and an individual is not subjected to the water for a long period.
4. Because of the large number of variables (i.e., ice water temperature, duration of therapy session, time lapse between the end of exercise session to the beginning of cryotherapy, etc.), much more research is needed in this area.

If your client takes an ice bath, it is important to know how much ice to use, and for how long, to avoid the risk of hypothermia. The goal is to have the water cold but tolerable. Filling up a bathtub with cold water and putting a tray of ice cubes in it is a good place to start, then adding more ice as needed. The water temperature should be between 55 and 60 degrees F and typically five to 10

minutes is a good duration. Once an ice bath is over, it is advised to allow the legs/body to warm up gradually, instead of using an external heat source.

MOVE AROUND ... A LITTLE

Depending on a runner's level of fatigue post-race, some walking around is helpful in the recovery process. However, it is ill advised to spend the remainder of the day after a race constantly walking around. This will likely hinder the recovery process and, more to the point, your client might not be able to do so. Therefore, it is probably not a good idea for your client to make plans that require lots of energy after a race. The longer the race, the more applicable this is.

Post-Race Assessment

Upon finishing a race, your client will likely be exhausted and feeling a wide range of emotions from elation to disappointment. Regardless of the emotional response, a self-assessment of one's performance directly after an event is not recommended. Your client should wait until having a chance to clean up, refuel, and relax before assessing how the race went.

Self-assessment will provide both you and the client with valuable information. The following areas represent common assessments:

- Areas in which the client felt prepared and underprepared
- Strengths and weaknesses
- Accuracy of pacing
- Fueling strategy

Odds and Ends

- Most hotels have a checkout time in the late morning. It is advisable for your client to try to get a late checkout or book the hotel for another day so as to have somewhere to get cleaned up and recover after the race.
- Please remember to tell your client to thank the event organizers and volunteers for all their hard work!
- Maintaining a proper diet and hydration levels during travel is important. Do not rely on fast food at airports and gas stations for fuel while traveling.

- Allow ample time for prerace activities. Your client does not want to be on such a tight schedule that a slight delay might negatively affect other aspects of race preparation.

❖ SUMMARY

- Before leaving for an event, your client must go through a checklist to make sure nothing is forgotten.
- If flying to an event, running shoes must be taken as a carry-on.
- If traveling to a popular event, it is advised to make lunch/dinner reservations ahead of time.
- Knowledge of a race course is extremely important.
- Once clients have their race program, they should create a time-based schedule to ensure they are where they need to be at certain times.
- In order to be comfortable racing in bad weather, one has to train in it.
- Your client must be aware of all the rules of the race.
- Nothing new on race day!
- The client should do some, but not too much, walking after the race.
- If a race has an athlete tracking system, your client should inform friends and family so they can track the runner during the race.
- Have clients set multiple alarms on race day to ensure they do not oversleep.
- Post-race fueling and care are critical to recovery.
- If staying in a hotel, it is advised to book a room for an additional day so clients have somewhere to clean up and rest after their event.
- Following are areas that must be focused on the week before a race:
 - Rest and recovery
 - Nutrition and hydration
 - Supplies
 - Confirm travel plans (if any)